

Worksheet 5A: Angles, Radians, and the Unit Circle — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Degrees and Radians

1. $\frac{\pi}{2}$
2. π
3. $\frac{\pi}{4}$
4. $\frac{\pi}{3}$
5. $\frac{\pi}{6}$
6. $\frac{2\pi}{3}$
7. $\frac{3\pi}{2}$
8. $\frac{7\pi}{4}$
9. $\frac{5\pi}{6}$
10. $\frac{5\pi}{4}$
11. 30°
12. 45°
13. 60°
14. 120°
15. 150°
16. 225°
17. 210°
18. 330°
19. ≈ 0.00175 rad

Part B: Arc Length

20. (a) 5π cm (b) 10π cm (c) 20π cm

21. (a) 6,371 km (b) ≈ 0.930 rad (c) $\approx 5,930$ km
22. (a) ≈ 0.1257 rad (b) $\approx 6,366$ km (c) Very close!
23. (a) ≈ 0.00908 rad (b) $\approx 3,490$ km (c) Very close!
24. $\approx 1,390,000$ km (actual $\approx 1,392,000$ km)
25. (a) ≈ 0.000242 rad (b) $\approx 142,000$ km

Part C: Unit Circle

26. 1, 0
27. $\frac{\sqrt{3}}{2}, \frac{1}{2}$
28. $\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}$
29. $\frac{1}{2}, \frac{\sqrt{3}}{2}$
30. 0, 1
31. $-\frac{1}{2}, \frac{\sqrt{3}}{2}$
32. $-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}$
33. $-\frac{\sqrt{3}}{2}, \frac{1}{2}$
34. -1, 0
35. $-\frac{\sqrt{3}}{2}, -\frac{1}{2}$
36. $-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$
37. $-\frac{1}{2}, -\frac{\sqrt{3}}{2}$
38. 0, -1
39. $\frac{1}{2}, -\frac{\sqrt{3}}{2}$
40. $\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$
41. $\frac{\sqrt{3}}{2}, -\frac{1}{2}$
42. $\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$
43. $\frac{\sqrt{3}}{2}, -\frac{1}{2}$
44. $\frac{\sqrt{3}}{2}, \frac{1}{2}$
45. -1, 0
46. 1
47. $\frac{1}{2} + \frac{1}{2} = 1$
48. $\frac{1}{4} + \frac{3}{4} = 1$
49. $\cos \theta = \frac{4}{5}$

Part D: Tangent and Applications

50. 0

51. $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

52. 1

53. $\sqrt{3}$

54. Undefined (division by zero)

55. $-\sqrt{3}$

56. -1

57. 0

58. (a) $\frac{\pi}{6}$ rad (b) $\frac{\pi}{3}$ rad or 60°

59. 90°

60. (a) $\frac{\sqrt{3}}{2} \approx 0.866$ (b) $\text{NA} = 0.866$

61. (a) 0.940 (b) $\text{NA} \approx 1.43$ (c) Higher refractive index allows larger effective angle

62. (a) ≈ 12.3 nm (b) Requires multiple patterning techniques

63. (a) ≈ 43 nm (b) Physics limits further shrinking with longer wavelengths
